LIGHTYEAR

MS93 Judge and execution environment qualification

Word edition

Status: Bounded reference qualification passed; development gates remain before another autonomous campaign. **Date:** 28 September 2026.

MS93 establishes that the test equipment can accept a known operations journey and reject four deliberate defects using real Oracle and PostgreSQL observations. It follows MS92's 40 halted trials. The reference is private test equipment, never an autonomous success and never a source of expected business outputs for the builder.

Reference and negative controls

The original reference derives from the passing MS91 journey, with 1.5-second observation windows for rollback and lock waits. Its declared shape is one invoice with one aggregate order-linked line, partial shipments, credit and reversal, rollback and caller-keyed retry, and controlled concurrency. The operator accepted this scope conditional on native qualification. The signed summary records that condition as satisfied; it does not claim a line-by-line human source audit.

Control	Native qualification result
Original positive operations reference	Complete combined judge passed on both engines
Wrong order quantity	Both engines rejected the quantity outcome
Missing invoice posting	Both rejected missing accounting entries
Incorrect payment link	Both rejected the document relationship
Leaked rolled-back draft	Both rejected a surviving rollback record

Each fault was introduced after successful native application execution on fresh test databases. The negative qualification requires the specific intended rejection in each lane, the expected combined result and complete cleanup. The rollback mutation creates a committed leaked draft; it does not disable database rollback. Four rejected defects do not prove that every possible defect will be detected.

Reliable results and isolated execution

Every subsequent attempt writes a structured result: passed, business-failure, execution-failure, judge-error or insufficient-evidence. An exception leaves inspectable private details instead of a missing gate file. Builders receive a separate structural projection, never arbitrary private judge prose or expected monetary values.

Two equipment failures remain visible

The first original-reference run failed when an Oracle sample lacked rollback counters. Observer v2 retained incomplete samples without accepting them as witnesses. The first isolated support-reference run completed both databases but retained insufficient-evidence: Oracle rolled back and began the retry between samples, without an observed no-transaction interval.

Observer v3 accepts an ended transaction followed by a different transaction ID only with the same session and serial, exactly one rollback, applied undo, no intervening commit and a prior observed write

binding. Ambiguous transitions still fail. A fresh isolated support reference passed on both engines. A separate disposable-workspace reassessment passed that positive control and rejected all four preserved native fault captures under judge v3. The negative captures were replayed, not re-executed; original files and verdicts remain unchanged.

Public tools and deterministic support

The application container mounts only candidate source, a public launcher and its output directory. The trusted collector retains private judge code and captures. The host stops the application before collecting its trace and writing the execution envelope.

The local MCP server provides public contracts, allowlisted hash-verified API signatures, structural checks, deterministic support and bounded offline compilation. Its compiler has no database network or private judge mount. Failed calls consume compilation slots, and the budget survives server restarts. The real stdio transport was exercised without an LLM invocation.

JourneySupport handles plain decimal serialization, trace insertion and replacement, typed posting access and transaction cleanup. It was compiled against the pinned application API and exercised in the composed native reference. Its posting guard does not prevent internal application costing or reposting effects. Separate stage checks and review of the changed support source remain open.

Purchasing feedback and release gates

Source investigation tied the purchasing cost queue, cost history and accounting history to native costing and matched-invoice reposting. The proposed rule checks document, client, organization, product and accounting relationships; preserves old rows; verifies balance and captured history identity; and retains every raw difference. It does not grant unrestricted access to three tables. A fresh purchasing reference passed the proposed complete judge on both engines, with human reference review still pending.

The development exporter replayed all 78 MS92 attempts and found supported structural feedback in 45. It covers compiler exceptions, imports, method visibility, signatures, source-supported Boolean mismatches and missing fields after successful execution. It no longer floods an interrupted journey with downstream missing-field diagnostics. That is feedback coverage, not evidence that the old candidates would repair successfully. Independently established footprint and declared trace-type feedback remain release gates.

Validation and cost

There are nine retained native equipment attempts: three passed, four produced the intended business-failure controls, one execution-failure and one insufficient-evidence. All cleanup receipts verified. Summed attempt time is 3,256.641 seconds, approximately 54.3 minutes including preparation and cleanup. Builder and analyst calls were zero. This is not an autonomous success rate. The focused regression suites passed 114 tests, including all five result classes, observer edge cases, public-tool boundaries and deterministic Java plumbing.

Provenance and next decision

The operations reference and public requirements inherit MS91. The support variant replaces routine plumbing without changing the intended business journey. The purchasing reference derives from a retained MS92 candidate. Observer changes respond to the two preserved equipment failures; exporter changes respond to MS92 compiler and trace failures. These are disclosed development lessons, not improvements made inside a frozen campaign.

Before another generation campaign: finish separate component checks and reference reviews; connect the controlled MCP transport to a separately versioned controller; complete permitted feedback; freeze the inputs, judge, controller and budget; then authorize a small pilot followed by ten

fresh generations. Keep unfamiliar evaluation scenarios separate. MS92's old call authorization is not reused. Organization API migration remains deferred, and no GCP resources were started.

References and evidence scope

- Signed nine-attempt status
- Original conditional reference acceptance
- Judge v3 control reassessment
- Compact results and retained capture scope
- Implementation and remaining gates

Git contains all nine signed plans, gates, receipts and cleanup records. Full raw captures and archived execution sources remain in the recorded local run directories; they are required for complete native-judge replay. Local signatures establish attribution and integrity, not independent attestation or full platform qualification.